

ABBA4 Implicit: Canonical Theory

One symmetric projection around an unprojected triple jump

1 Unprojected fourth-order base

Let $\mathcal{A}_{q,t}$ be one signed endpoint-time ABBA map on two copies, with no intermediate diagonal projection. Define

$$\gamma = \frac{1}{2 - \sqrt[3]{2}}, \quad \delta = -\frac{\sqrt[3]{2}}{2 - \sqrt[3]{2}}.$$

The complete base map is

$$\mathcal{C}_{h,t} = \mathcal{A}_{\gamma h, t + (\gamma + \delta)h} \circ \mathcal{A}_{\delta h, t + \gamma h} \circ \mathcal{A}_{\gamma h, t}.$$

Because $2\gamma + \delta = 1$ and $2\gamma^3 + \delta^3 = 0$, the palindromic composition cancels the leading third-order defect of a symmetric second-order base. The middle duration is negative, and the two copies pass continuously through all three factors.

2 One outer symmetric projection

For

$$E = \begin{pmatrix} I \\ I \end{pmatrix}, \quad P = \frac{1}{2} \begin{pmatrix} I & I \end{pmatrix}, \quad G = \begin{pmatrix} I & -I \end{pmatrix}, \quad N = G^T,$$

the reduced formulation is

$$M(\mu) = \mathcal{C}_{h,t}(Ez + N\mu), \tag{1}$$

$$r(\mu) = GM(\mu) + 2\mu = 0, \tag{2}$$

$$z^+ = P(M(\mu) + N\mu). \tag{3}$$

If J_j is the Jacobian of signed factor j , then $J_{\mathcal{C}} = J_3 J_2 J_1$ in execution order and

$$D_{\mu}r = GJ_{\mathcal{C}}N + 2I.$$

Every residual evaluation traverses all three ABBA factors.

3 Configurations

The sole implementation of `ABBA4Implicit` exposes two projection formulations, Newton or Broyden, physical or fully extended state, and optional physical energy tracking. Fully extended execution always enables energy evolution. The selected values are global to the outer step. In the simultaneous formulation the corrected doubled output joins μ in the unknown; elimination gives the reduced residual above. State extension changes what is duplicated, while physical tracking adds only a triangular $\kappa = k/2$ sidecar. Neither choice introduces intermediate projection.

4 Removed per-map implementation

The current default uses one outer projection. The former `after_each_abba_map` implementation, with three distinct roots and intermediate diagonal re-embedding, has been removed. It defined a different numerical map, since projection and factor evolution do not commute. An explicit request for that placement now raises an error. The legacy single-projection constructor is a compatibility factory for the current default and does not define another method.

5 Properties and limitations

The ideal method is symmetric and designed for order four. Geometric claims refer to the converged projection root; finite nonlinear tolerances perturb the map. Negative internal time increments require the potential and dynamics to be evaluable slightly outside the outer interval. Observer records preserve the three ordered factors so the accepted-map tangent can be reconstructed.

6 Implementation correspondence

The public `ABBA4Implicit` configuration is in `src/simulation/methods/abba/order4_implicit.py`; the exterior-projection residual and numerical kernel are in `src/simulation/methods/abba/projection_outer.py`. The older single-projection module provides compatibility imports only. The common ABBA stages and projection helpers live in the neighboring private modules. The canonical architecture document `../simulation/abba4-implicit-simulation-architecture.md` gives the dimensions, diagnostics, and the common prepared integration lifecycle.

7 Explicit residual formulations and ideal-root tangent

Let $w = C(Ez + N\mu)$ and let a, b be the corrected final copies. The simultaneous unknown is (a, b, μ) and its equations are

$$F(a, b, \mu; z) = \begin{pmatrix} a - w_u - \mu \\ b - w_v + \mu \\ a - b \end{pmatrix} = 0.$$

Eliminating a, b gives $GC(Ez + N\mu) + 2\mu = 0$. Thus both formulations define the same ideal map; their finite-precision solve paths and work may differ. Newton uses the exact block derivative, while Broyden updates an approximate residual Jacobian. Neither method projects the intermediate factors.

At a converged reduced root, set

$$J_C = J_3 J_2 J_1, \quad K = G J_C N + 2I, \quad L = G J_C E.$$

For nonsingular K , the implicit function theorem gives

$$D_z \mu = -K^{-1} L, \quad D_z z^+ = P J_C (E - N K^{-1} L).$$

The inverse denotes a linear solve in the implementation. This tangent is one outer implicit differentiation. A product of three independently projected physical tangents would describe the removed implementation instead. For independent particles, physical Jacobian blocks can be evaluated per particle, preserving the component-major physical layout.

8 State and energy strategies

For N physical GC particles the accepted physical state has $2N$ components. Physical tracking appends N auxiliary values $\kappa = k/2$ after the physical solve, using the signed accepted stage traces; it does not change the multiplier root. Fully extended execution currently supports one GC particle and duplicates $Z = (x, y, t, k)$, using an eight-component base and a four-component accepted state. The full time is synchronized at outer boundaries; energy evolution is enabled.

Strategy	Base size	Reduced unknown	Simultaneous unknown
Physical, tracking off or on	$4N$	$2N$	$6N$
Fully extended, one particle	8	4	12

There are three normalized state/energy strategies, two formulations and two nonlinear solvers: $3 \cdot 2 \cdot 2 = 12$ configurations. With energy enabled, the configuration comparison has $2 \cdot 2 \cdot 2 = 8$ combinations.

9 Method instances and diagnostic meaning

The public class inherits initialization, advance, observation and output extraction from the common ABBA numerical class. Inherited `IntegrationMethod.integrate` creates a fresh instance of the same concrete class for each execution, then uses the common driver. The instance owns its state policy, projection equation and coefficients; no separate context or callback-based method record is needed. Each advance returns accepted state, small statistics and the typed trace of its single outer projection.

The common fixed controller owns the uniform accepted grid and independent shortened output advances. The common coordinator owns progress, collection, optional observation and final output. The collector stores requested samples and accepted metric rows, without retaining stage records by default. Export extracts physical coordinates and auxiliary energy, never another numerical projection.

One ABBA4 step has one accepted nonlinear solve and three unprojected ABBA factors per residual evaluation. Substep metric arrays consequently have one column; physical observation records have three unprojected factor snapshots and one multiplier. Full observation has one composed-base record and its full tangent. Observer and shadow work do not alter main-step counters.

10 Verification and historical notes

The current implementation is checked for fourth-order convergence, reversibility, multiplier scaling, exact-root tangents, solver/formulation agreement and observer isolation. Architectural regression compares against the retained outer-projection implementation, not against the removed map. Nonlinear tolerances and floating-point precision limit formal properties; trajectory accuracy requires refinement, independently of energy diagnostics.

The per-map simultaneous-formulation and Jacobian companions are retained as historical derivations for interpreting earlier data. The active architecture is the six-phase diagram in the model's `simulation/` directory.